



FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

Airflow Setup - First Use

Windows/WSL2 beginner path; install only when C3-03-L02 begins.

Artifact	C3-039
Status	2026 - LEARNER EDITION
Phase	C3-03 Orchestration & Reliable Workflows 13 lessons 34-46 focused hours
Prerequisite	C3-02 Gate PASS; C2B Python/quality/ownership foundations assumed, orchestration taught here.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: simple mental picture -> tiny example -> guided Airflow/tool workflow -> break/fix -> independent changed task -> validation -> production reasoning -> evidence. A green scheduler box alone never proves correct data.



C3-03 Airflow local-learning setup

Use WSL2. Airflow 3.3.1 officially supports Python 3.10-3.14 and POSIX systems; Windows learners should use WSL2 or Linux containers. This course uses WSL2 + a dedicated venv so generated Airflow state stays isolated.

1. Open Ubuntu/WSL and make the project

```
mkdir -p ~/course3/orchestration/c3_03
cd ~/course3/orchestration/c3_03
python3 --version
```

Expected: Python reports a supported 3.10-3.14 version. If it does not, repair Python/WSL before installing Airflow.

2. Create an Airflow-specific virtual environment

```
python3 -m venv .venv
source .venv/bin/activate
python --version
python -m pip install --upgrade pip
```

3. Reproducible Airflow install

```
AIRFLOW_VERSION=3.3.1
PYTHON_VERSION="$(python -c 'import sys; print(f\"{{sys.version_info.major}}.{{sys.version_info.minor}}\")')"
CONSTRAINT_URL="https://raw.githubusercontent.com/apache/airflow/constraints-{{
AIRFLOW_VERSION}}/constraints-{{PYTHON_VERSION}}.txt"
pip install "apache-airflow=={{AIRFLOW_VERSION}}" --constraint "{{CONSTRAINT_URL}}"
pip check
airflow version
```

Expected: `airflow version` -> 3.3.1 and `pip check` reports no broken requirements.

4. Start only the learning profile

```
export AIRFLOW_HOME="$HOME/course3/orchestration/c3_03/.airflow"
airflow standalone
```

Keep this terminal open. Open the local URL shown by Airflow in your browser. Never copy the generated admin password into Git/evidence.

5. Discovery check

```
source .venv/bin/activate
export AIRFLOW_HOME="$HOME/course3/orchestration/c3_03/.airflow"
airflow dags list
```

Expected: the learner DAG appears after you place its source in the configured DAG folder. If not, inspect import errors before editing business logic.

Laptop safety

Use the AIRFLOW profile: WSL2 + Airflow + editor/browser. Keep Docker, Kafka and Spark closed unless a specific integration exercise requires them. Airflow documents a minimum around 4 GB, but actual needs depend on deployment.